

# ECE 447

## Lesson 18

### Signal sampling and reconstruction

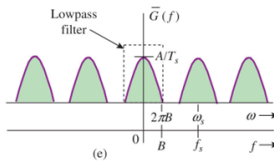
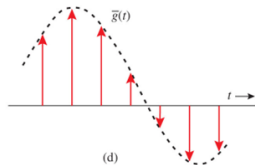
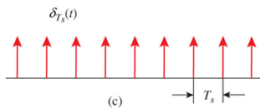
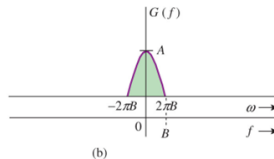
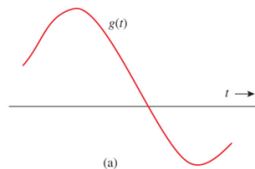


UNITED STATES  
AIR FORCE  
ACADEMY

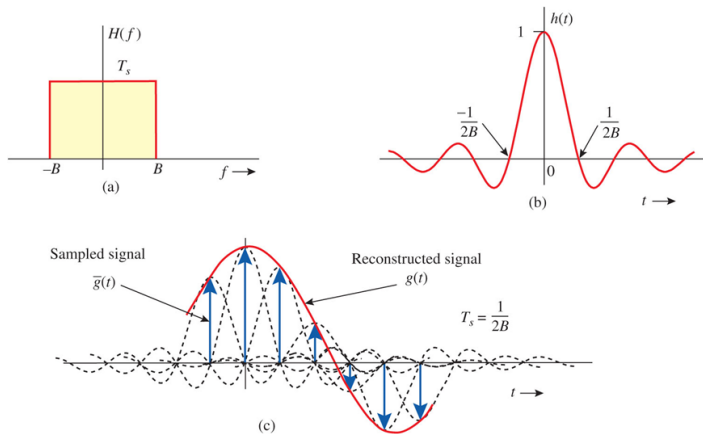
# SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
  - **HW4.** Assigned today. Due Lesson 21 (Tue, 6 Oct) to Gradescope.
  - **Lab 4.** Due Lesson 19 (Wed, 30 Sep) to Gradescope.
  - Everything from today through Lesson 32 is GR2 material (Chapters 5-8). GR2 is Lesson 35 (Tue, 17 Nov).

# SAMPLING

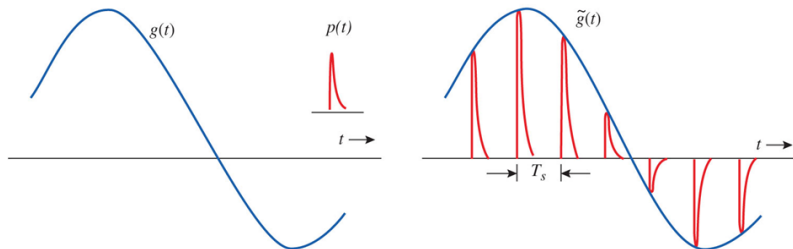


# IDEAL RECONSTRUCTION



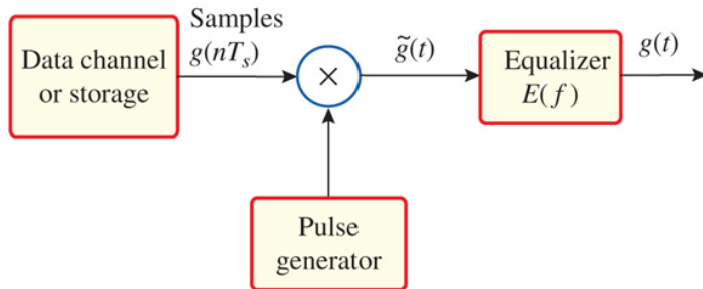
# NONIDEAL RECONSTRUCTION (INTERPOLATION)

- Interpolation pulse  $p(t)$
- $\tilde{g}(t) = p(t) * \bar{g}(t)$
- $\tilde{G}(f) = P(f)\bar{G}(f) = P(f)\frac{1}{T_s} \sum_n G(f - nf_s)$



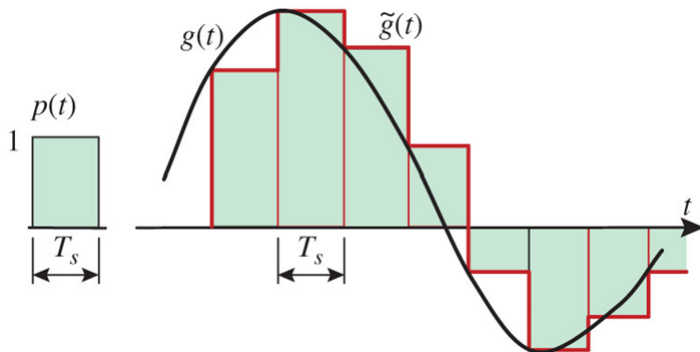
# EQUALIZERS

- $P(f)$  filters sampled signal - need to reverse filtering to obtain original
- Use equalizers!



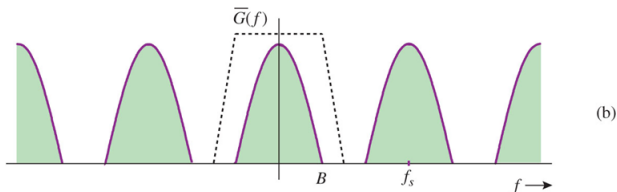
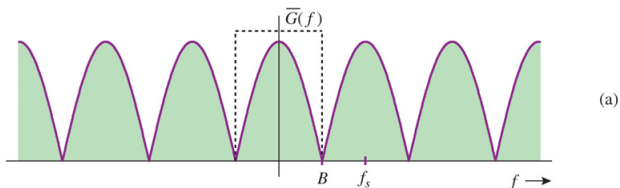
# INTERPOLATOR EXAMPLE - FLAT TOP PULSES

- Also known as a zero-order-hold pulse
- Samples signal and holds that value for  $T_s$

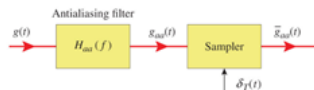
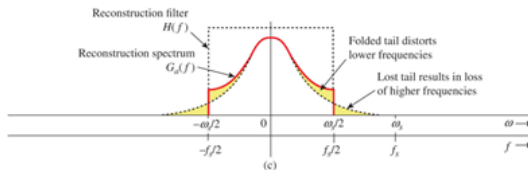
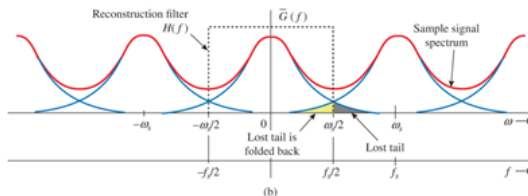
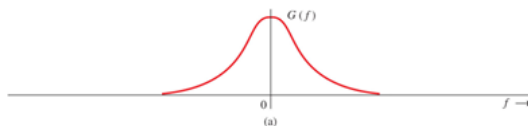


# REALIZABLE RECONSTRUCTION FILTERS

- Can't use ideal LPFs
- Need space between samples in spectra



# ALIASING



# EXAMPLE PROBLEM - 5.1-7